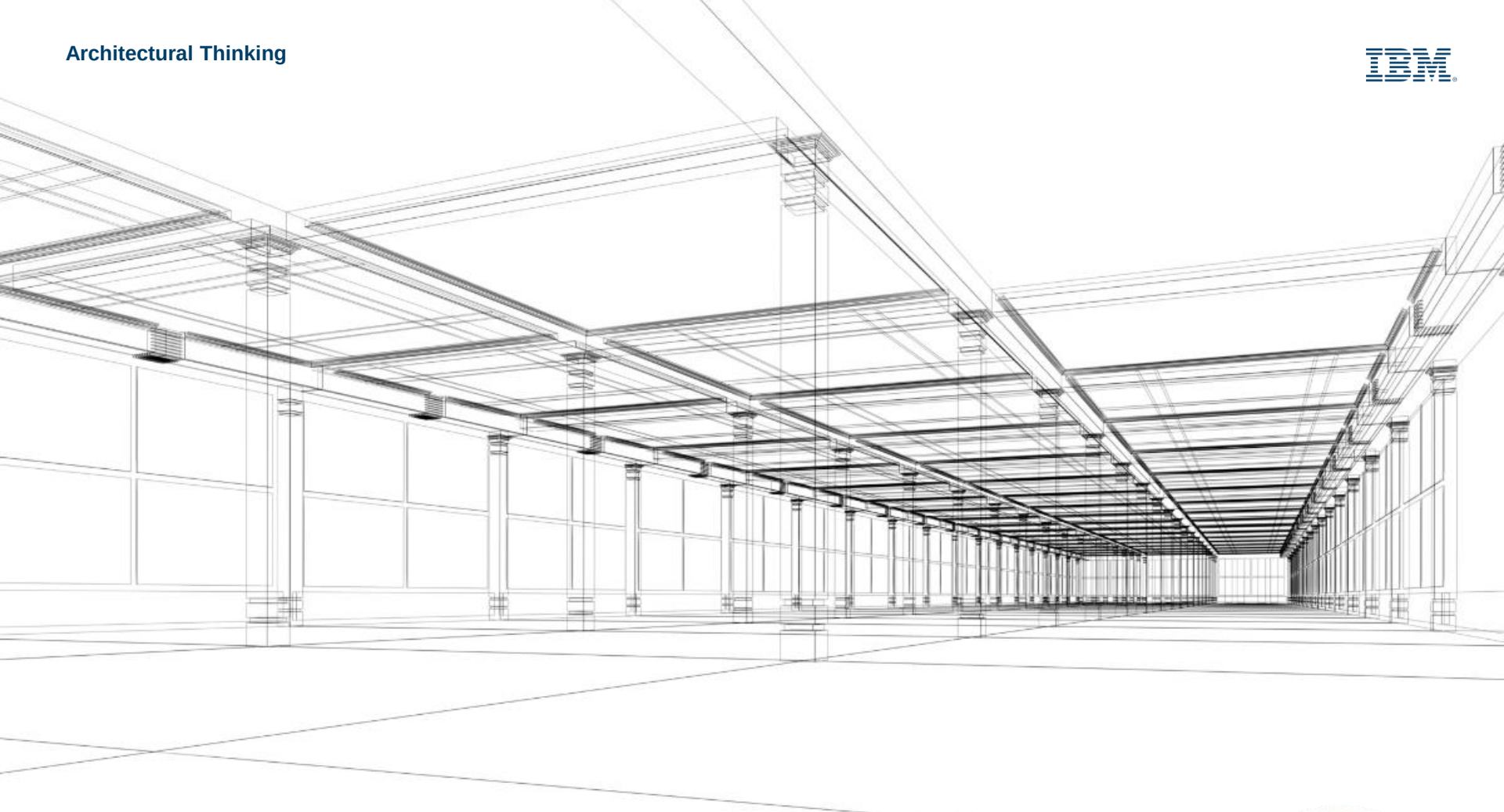




Tooling for IBM's Architects



Module outline



► Introduction and objectives

Project context

Architecture content

Information sharing

Other key tools for architects



- Learn about different categories of tools.
- Describe which tools make sense for which situation or phase of the project.
- Discuss sharing of information between teams, architects, and tools.

Artifacts usually fall into one of two categories: architecture content or project context



- This course has covered a core set of artifacts that contain the architecture content, for example:
 - Architecture overview
 - Component model
 - Architecture viability assessment
- Other documents may be inputs to or constraints on the architecture, and provide the context in which the architecture must live, for example:
 - IT governance (from the organization's Enterprise Architecture group)
 - IT standards (from EA)
 - Business strategy (from the Business Architecture team)
 - Application portfolio (from BA)
 - Project workplan (from the Project Management office)

Module outline



Introduction and objectives

► **Project context**

Architecture content

Information sharing

Many tools provide the Architect with a means of understanding and managing the context in which he or she is working



Architects don't just produce architecture artifacts. Architects work on, contribute to, or use information associated with dozens of *contextual* artifacts throughout the life cycle of their work.

- Pre-project or bid and proposal
 - Inventory analysis
 - Portfolio planning
- Project kickoff
 - Estimation
 - Proposal development
- Project kick off
 - Selection and customization of engagement process, methods, techniques
- Project execution
 - Work planning and management
 - Governance
 - Collaboration and communication
 - Asset management
 - Reporting and tracking
 - Standards



Project management tooling examples:

- Microsoft® Project
- ProjectLibre
- Planner (Project Management Software)
- Primavera (Oracle)
- Rational Team Concert™ for light agile project plans
- ZenHub

Method content management tooling examples:

- Rational Method Composer
- Rational Software Architect extension for integrated architecture frameworks
- Wikis and IBM Connections Communities

Governance tooling examples:

- Unicom Focal Point™ for graphs depicting priority and strategy of project.
- Unicom System Architect for models depicting strategy, goals, and objectives.
 - An older version, Rational System Architect, is available for free use within IBM
- GS Risk (PC@IBM)
- Risk management software included in PM tools or separate portfolio management tools.

Module outline



Introduction and objectives

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► **Architecture content**

Information sharing



- Rapid solution creation tools
 - Business stakeholders may require a rapid turnaround on an initial and high-level view of the system to approve sponsorship.
 - Technical stakeholders may need an early view of the system to determine implications related to their domain prior to investing in detailed work product development.
 - Some work products will only have a short period of relevance to members of an initiative and will not require depth or maintenance.
 - Each artifact may be produced using a different stand-alone tool.
- Integrated tools
 - For similar purposes as ad hoc, but with a greater degree of formality, depth, and detail.
 - May facilitate some sharing between tools.
- Comprehensive multi-team tools
 - Where a single tool is desirable to be used to produce the majority of work products and enable a holistic view across work products.
- Special purpose tools which may or may not integrate

The 15 work products addressed by Architectural thinking might have to be rendered or sketched quickly and with limited formality



Rapid solution creation tools may include:

- IBM Rational DOORS Next Generation.
 - Capture
 - User Stories and Use Cases
 - Requirements
 - Business Processes
- Microsoft PowerPoint, Word, Excel.
- Visio (diagrams can be imported into Rational System Architect)

Comprehensive tools offer many features to enable a holistic view of the architecture



Salient features:

- Ability to model an element only once and have all related work products leverage it to ensure consistency.
- Ability to manage the dependencies among work products using impact reports.
- Wide diversity of models based on modeling standards.
- Integrated with tools used by other stakeholders to exchange content upstream or downstream.

Candidate tool sets:

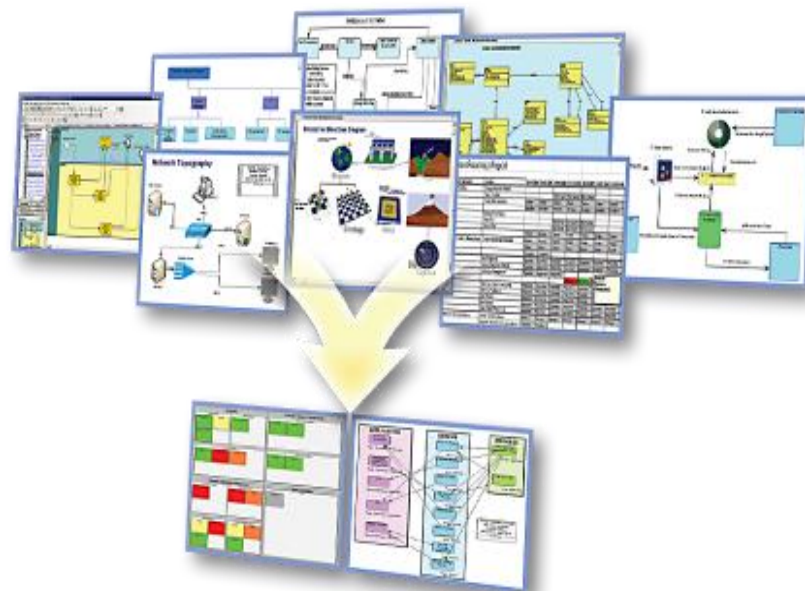
- Rational System Architect by IBM (now owned by UNICOM Systems)
- SA4TeamSD extension to rational system architect provides direct input using a work product dependency diagram as a graphical navigator.
- ARIS by Software AG/IDS Scheer.
- Enterprise Architect by Sparx systems.
- Archi – The Free ArchiMate Modeling Tool
- Cognitive Architect (Q4 2017)

A focus on Rational System Architect as a comprehensive solution development tool



Supports a wide range of work products for a complete system view

- Link, consolidate, and analyze architecture information in blueprints
 - Strategies, goals, and organization
 - Applications, services, and data
 - Systems, networks, and components
- Simplified modeling with robust support for standards
 - Create a common and consistent information resource to guide enterprise planning
 - Multiple modeling standards in one tool
 - Changes are propagated throughout the data blueprint, simplifying maintenance
- Meta-model applied at encyclopedia (asset) level determines the structure of model and available views
 - TOGAF
 - TeamSD (SA4TeamSD)
 - ...



Rational system architect integrations include:

DOORS, DOORS Next Generation, focal point, websphere business modeler, rational software architect and asset manager, Tivoli TADDM and CCMDDB, and InfoSphere data architect.



Use case model

- IBM Rational DOORS Next Generation manages the high-level use case diagrams along with text and nonfunctional requirements. This tool is best targeted for standard IT and highly collaborative environments.
- Use case diagrams can be built and managed in Rational Software Architect (via UML-2)

Nonfunctional requirements

- Rational DOORS supports rigorous requirements management for engineering and compliance environments. It provides some use case diagram support along with nonfunctional requirements and standards management.

System context diagram, principles, architecture decisions

- While there are no highly specialized tools that are required to detail these work products individually, rational system architect is highly effective for general diagramming.



Viability assessment

- This type of work product could reasonably be captured in almost any tool.
 - The TeamSD template uses Microsoft Word.
 - Because Rational Quality Manager facilitates risk-based testing in test plan creation to help testers prioritize their efforts based on probable concerns with high impacts, it could be helpful to share the viability assessment with testing teams that use this tool.

Validation

- DOORS supports rigorous requirements management for engineering and compliance environments. It provides some use case diagram support along with nonfunctional requirements and standards management.
- Architects can use the automatic validation features in their modeling tools to ensure that standards are followed and to test their designs directly.
- Architects can view Rational Quality Manager to ensure that the validation test plans and test motivators accurately depict what is needed to validate the system.
 - Test analysts will use the RQM Manual Tester, Rational Functional Tester, and Rational Performance Tester to conduct much of the validation.



Data modeling

- InfoSphere Data Architect is a collaborative enterprise data modeling and integration design tool to:
 - discover, model, relate, and standardize diverse and distributed data assets.
 - Enforce model conformance with enterprise standards.
 - Leverage industry data models for best practices.
- InfoSphere Data Architect integrates with Rational Software Architect to ensure consistency between the application models and the data models.

Component and service models and deployment units

- Rational Software Architect can define the executable and reusable aspects of a software system and interface with source code files.
- Use Rational Software Architect to specify the components and their interfaces and automatically generate the source code in C++, Java® SE, and XML, in addition to WebSphere target architecture.
- Rational Software Architect can model all four types of deployment units to present the logical or operational layout of the application and specify details about the runtime behavior of an application component.

Operational model

- Rational System Architect enables solution architects to create and manage operational models most efficiently by leveraging many model elements previously created and by automatically reflecting their changes.



- A new internal architecture tool is under development and expected to have initial release at the start of Q4 2017.
- Focus on two key elements of Solution Design
 - Finding Candidate Assets for direct reuse
 - Lightweight architecture design authoring with integrated meta-model

The screenshot displays the Cognitive Architect web application. The top navigation bar includes the title 'Cognitive Architect', a 'Workspace' button, a search bar, and a user profile icon. Below this, tabs for 'Private', 'Collaboration', and 'Bookmarks' are visible. The main content area shows a grid of architectural solutions. A large central card features a blue plus sign and the text 'Add Architectural Solution'. To its right are two smaller cards: 'CAC Blockchain Reference Architecture - v20170725' (labeled as a Reference Architecture with tags for Public cloud, Blockchain, and Microservices) and 'Cognitive Conversation Runtime Architecture' (labeled as a Solution Architecture with a Cognitive tag). On the left, a sidebar menu lists the navigation structure: 1. Overview, 2. Scope (with sub-items 2.1 Business Challenge and 2.2 System Context), 3. Requirements (with sub-items 3.1 Use Cases, 3.2 Functional Requirements, and 3.3 NFRs), and 4. Architecture (with sub-items 4.1 Architecture Overview, 4.1.1 Services View, 4.1.2 Enterprise View, 4.1.3 IT System View, and 4.2 Architecture Decisions). The '4.1.3 IT System View' is currently selected, displaying a detailed architectural diagram. This diagram illustrates the integration of 'Public Net' (containing Conversational Endpoints, Public APIs, and 3rd Party APIs) with 'Edge Services', 'Application Logic', 'Conversation' components, 'Answer Storage', 'Discovery', and 'Enterprise Systems' (including Business Process and Enterprise APIs/Services). A 'Description' panel on the right provides details for the selected view: 'Full-function Enterprise Service Bus with message handling, publish/subscribe and complex event processing capabilities.' and identifies the 'Type' as 'AAD IT System View'. The IBM logo is positioned in the bottom right corner.



Business component and process modeling

- Component Business Model (CBM) tool by GBS
- IBM BPM, WebSphere® Business Process Modeler (BPM), or Blueworks Live™
 - WebSphere Business Process Modeler integrates with Rational Software Architect to help ensure that Architects align business services with business processes.
- IBM Rational DOORS Next Generation (both BPM and storyboard)

Information modeling

- Data Architects use IBM InfoSphere® Metadata Workbench to access and report on operational metadata to understand the impact of making changes to the information environment, helping solution architects avoid potentially disruptive modifications to existing processes.
- Data Architects use IBM InfoSphere Discovery to automatically identify and define data relationships and business objects to prepare a solution architect on an information-centric project like MDM

Module outline



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► **Information sharing**



Sharing among tools

- IBM BPM, WebSphere® Business Process Modeler (BPM), or Blueworks Live™
- From “upstream” tools used by business architects and enterprise architects.
- To tools used by solution architects.
- To “downstream” tools used by designers, developers, and testers.
- To formalize a rapid tooling-developed sketch by importing into an integrated tool.
- To trace requirements end to end in the project life cycle.
- To share solution information with systems operation or reporting tools.

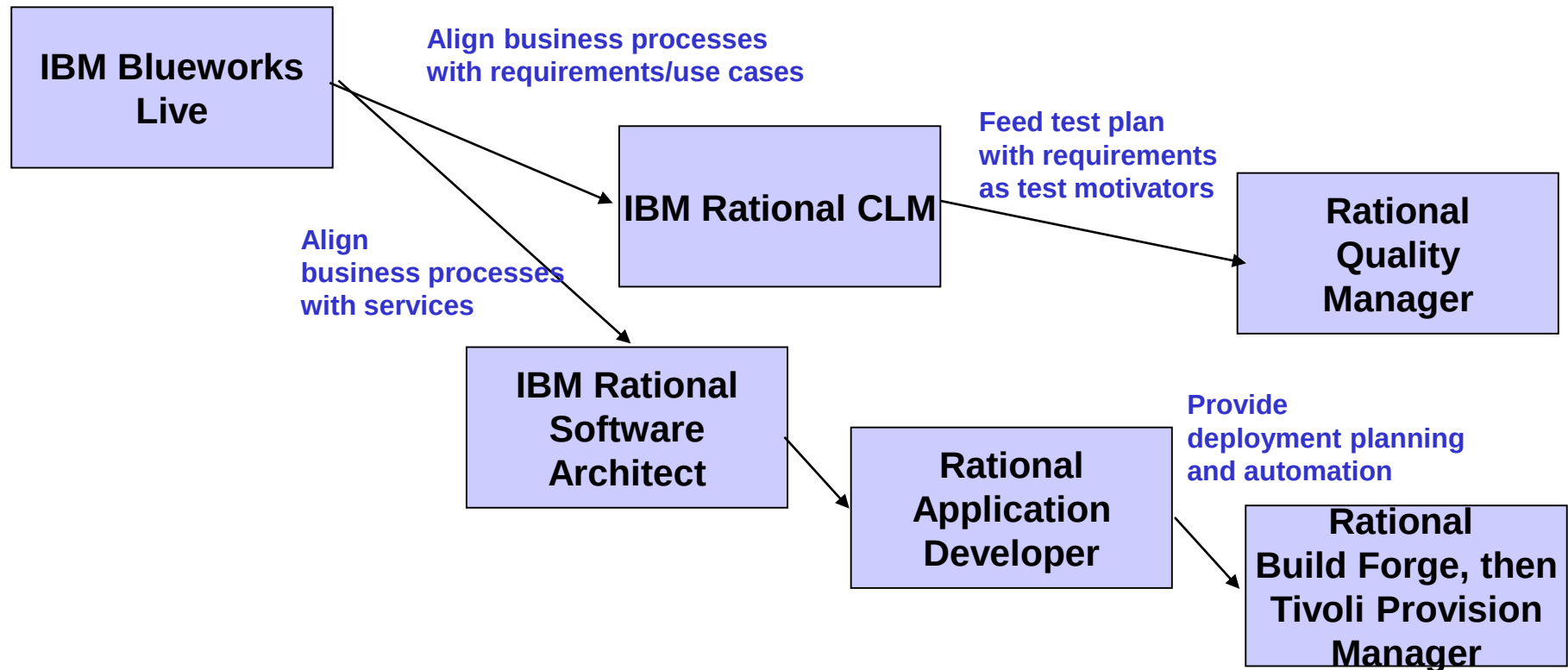
Sharing among stakeholders

- There are challenges with sharing between disparate communities with divergent skills and using different tools.
- A federated team may work better in some cases (separate models).
- A tightly integrated team may work better in others (one model)

Sample information flow among tools (Rational example)



The value provided by architecture and other tooling is demonstrated nowhere so much as in the enablement of the exchange of information among tools.





Federated team

- Rational Software Architect for Design Manager
 - Have a central location for your organization to collaborate on software and systems designs.
 - Provide easy access for stakeholders to designs through a web client.
 - Conduct design reviews that allow stakeholders to provide comments and visual markups linked to the designs.
 - Visualize relationships between designs and life cycle artifacts through a dependency relationship diagram. Sharing among stakeholders

Tightly integrated team

- Rational Software Architect Model Management
 - Rational Team Concert, ClearCase®, and GitHub are able to support the configuration and change management of a model developed by many on a team.

Module outline



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Information sharing

► Other key tools for architects

The Architect is involved in many supporting tasks requiring tooling



- Understanding the Customers and Opportunities
 - Engage, Client Expansion and Client Vantage, all part of Client Engagement Acceleration
 - Various aggregator applications, e.g., Bluemine, FirstRain, Crystal+
 - Sales Connect - to track opportunities.
- Asset Management Tools
 - General asset repositories:
 - Box
 - GitHub Enterprise
 - IBM Connections Files
 - Curated Search
 - IBM Solution Gateway
 - iGBS
 - Solution Seeker
- General Collaboration Tools
 - Slack
 - GitHub Enterprise
 - Sametime
 - Mural

The Architect is involved in many supporting tasks requiring tooling

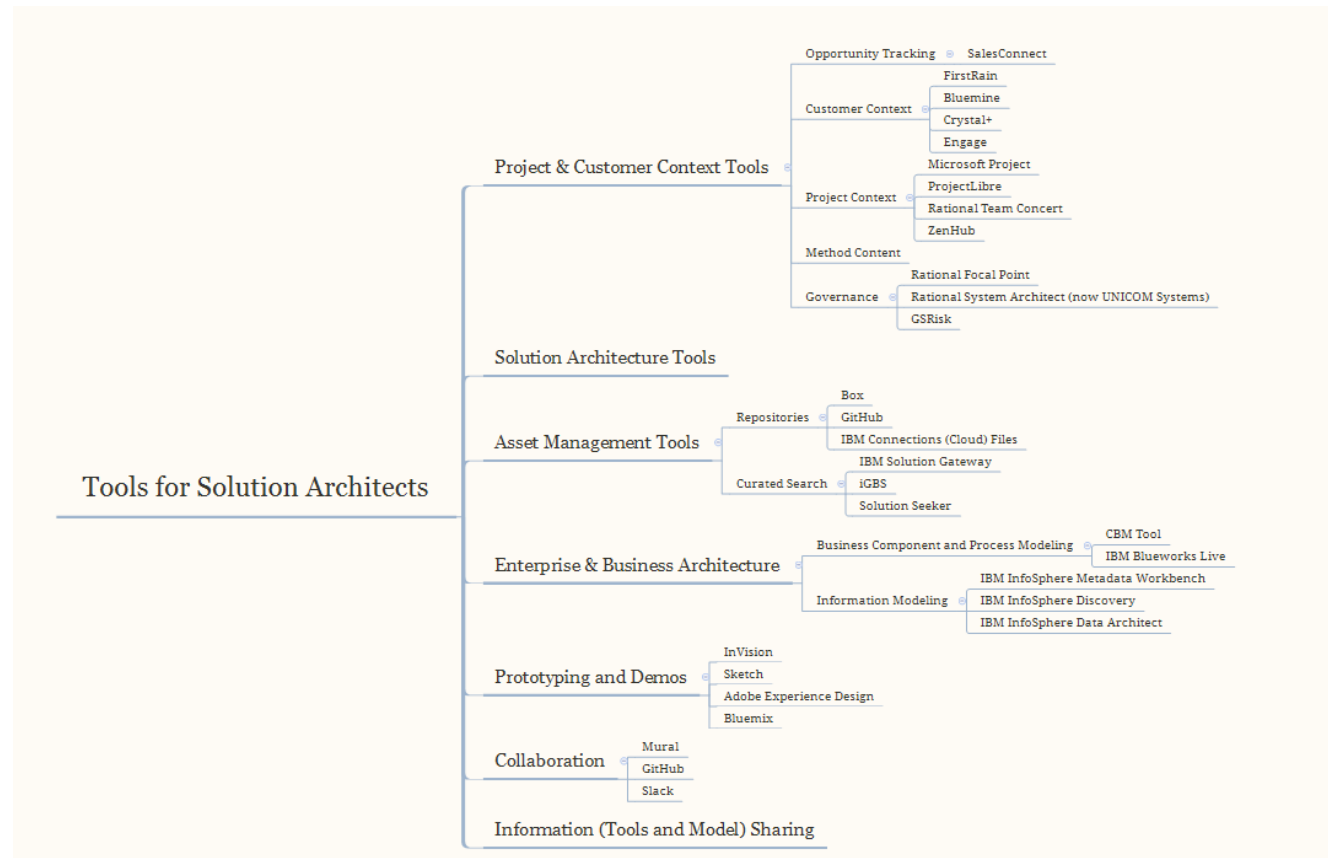


- Prototyping and Demo –
 - Building prototypes and demos are important to gain understanding and buy-in from Stakeholders
 - An important part of playbacks in IBM Design Thinking
- Build mid to high resolution wireframes:
 - Sketch
 - Adobe Experience Design
 - Adobe Illustrator
- Animate wireframes to build prototypes
 - InVision
 - Adobe Experience Design
- Rapidly build demos using PaaS platform
 - Bluemix
- Virtualization support
 - VMware Workstation / Fusion
 - Parallels Desktop
 - Zen
 - Docker



References for this chapter are posted to a box folder

Tools for Architects



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